Catastrophic Forgetting in Foundation Model Fine-Tuning: Cross-Domain Evidence from Time-Series Forecasting and IoT Anomaly Detection

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April 24, 2026

Abstract

Fine-tuning foundation time-series models destroys the pre-trained features that make them effective. We demonstrate this *catastrophic forgetting* mechanism across two domains with controlled experiments on Moirai [Woo et al.(2024)] (13.8M parameters).

On **ETTh2 forecasting**—where Moirai zero-shot outperforms Linear baselines by 28–45%—fine-tuning with NLL or NLL+SupCon loss degrades MSE by 11–16% after 20 epochs (10 seeds), while CKA between pre-trained and fine-tuned representations drops to 0.937–0.964. Freezing the encoder eliminates forgetting (CKA=1.000, <1% MSE change).

On **IoT anomaly detection**—using stealth-controlled synthetic attacks with protocol constraints (Modbus, MQTT, CoAP)—the same forgetting pattern emerges: unfrozen fine-tuning collapses Moirai to from-scratch CNN level (AUC≈0.56), while freezing recovers AUC to 0.606. A CNN trained with the same distributional NLL loss (AUC=0.598±0.042) rules out the loss-function confound, confirming that architecture-specific forgetting—not the training objective—drives the degradation.

Three findings generalise across domains: (1) contrastive loss does not prevent forgetting; (2) forgetting scales with task complexity (longer horizons, higher stealth); (3) encoder freezing is an effective but incomplete mitigation. We release code, datasets, and calibration scripts.

1 Introduction

Foundation models for time series—Moirai [Woo et al.(2024)], Chronos [Ansari et al.(2024)], TimesFM [Das et al.(2024)], MOMENT [Goswami et al.(2024)]—are trained on billions of time points and achieve strong zero-shot performance on forecasting benchmarks. A natural next step is to fine-tune these models on domain-specific tasks: anomaly detection, classification, or short-horizon forecasting with limited labelled data.

The forgetting problem. We show that standard fine-tuning of Moirai systematically destroys the pre-trained features that make it effective—a form of catastrophic forgetting [Kirkpatrick et al.(2017)Kirkpatrick, Pascanu, Rabinowit that occurs even when the training loss decreases monotonically. On ETTh2 forecasting, where zero-shot Moirai outperforms a Linear baseline by 28–45%, fine-tuning with NLL loss degrades MSE by 11–16% after 20 epochs (10 seeds). On IoT anomaly detection, unfrozen fine-tuning collapses Moirai to from-scratch CNN level. In both domains, freezing the encoder eliminates forgetting (CKA=1.000, near-zero performance loss).

Why this matters. Catastrophic forgetting is well-studied in continual learning [Kirkpatrick et al.(2017)Kirkpatrick, Pascanu, Ra Li and Hoiem(2018)] and NLP fine-tuning, but less characterised for time-series foundation models. Unlike language models, where forgetting manifests as degraded generation quality, time-series forgetting directly harms the distributional predictions that underpin anomaly scoring and probabilistic forecasting. Our cross-domain diagnosis provides controlled evidence that this is a general mechanism, not a property of a specific task or loss function.

Experimental approach. We design a controlled forgetting diagnosis with three conditions: (B) full fine-tuning with NLL, (C) NLL+supervised contrastive loss, and (D) frozen encoder. We deploy this on two domains: (1) ETTh2 forecasting [Zhou et al.(2021)Zhou, Zhang, Peng, Zhang, Li, Xiong, and Zhang], chosen via a *pre-training value gate* (zero-shot must exceed Linear by $>20\%$) to ensure pre-trained features are demonstrably useful; and (2) IoT anomaly detection using HNIDS (**H**ard-**N**egative **IDS**), a framework that synthesises stealth-controlled attacks via closed-form perturbations with protocol constraints (Modbus, MQTT, CoAP). Both domains use CKA (Centered Kernel Alignment) and weight drift as forgetting diagnostics.

Contributions. We make three contributions:

1. **Cross-Domain Forgetting Diagnosis.** Fine-tuning Moirai destroys pre-trained features on ETTh2 (10–16% MSE degradation, CKA 0.937–0.964) and IoT anomaly detection (7–15% AUC drop). Freezing the encoder eliminates forgetting in both domains (CKA=1.000). Contrastive loss does not prevent forgetting.
2. **Controlled Experimental Design.** A CNN-NLL control (AUC=0.598 ± 0.042) resolves the loss-function confound between MSE-trained CNN and NLL-trained Moirai. All comparisons use 10-seed bootstrap CIs with Cohen's d and p-values. A pre-training value gate on ETTh2 establishes that features are genuinely useful before testing whether fine-tuning destroys them.
3. **Stealth-Attack Benchmark.** Four IoT attack archetypes with closed-form definitions at three stealth levels (Eqs. 2–5), protocol validation (Modbus, MQTT, CoAP), and leave-one-out generalisation (mean +0.186 F1). Code and datasets at `https://github.com/anupamme/iotsf_demo`.

Paper organisation. Section 2 reviews related work. Section 3 formalises the problem. Section 4 describes HNIDS. Section 5 presents the ETTh2 forgetting diagnosis. Section 6 replicates on IoT anomaly detection. Section 7 provides cross-domain analysis. Section 8 concludes.

2 Related Work

Traditional IoT IDS. Statistical methods (Z-score, IQR) and Isolation Forest [Liu et al.(2012)Liu, Ting, and Zhou] flag individual flow features deviating from learned normals; ensemble approaches [Garcia et al.(2014)] improve precision. All share a weakness: calibration on known, high-magnitude attacks and failure on stealthy perturbations. We use the CICIoT2023 dataset [Neto et al.(2023)] throughout.

Deep-learning anomaly detection. USAD [Audibert et al.(2020)] uses adversarially trained dual-autoencoders. TranAD [Tuli et al.(2022)] applies context-aware Transformers. The Anomaly Transformer [Xu et al.(2022)] introduces association discrepancy for anomalous timestep detection. PatchTST [Nie et al.(2023)] captures long-range dependencies via patch-based Transformers. We include all four as baselines.

Generative augmentation for security. GANs [Lin et al.(2022)] suffer mode collapse; TimeGAN [Yoon et al.(2019)Yoon, Jarret improves temporal fidelity; Diffusion-TS [Zhang et al.(2024)] achieves state-of-the-art quality. Ours is the first to apply hard-negative contrastive augmentation for IoT IDS with protocol constraints, using closed-form perturbations (Diffusion-TS backbone deferred to future work).

Adversarial training. FGSM [Goodfellow et al.(2015)Goodfellow, Shlens, and Szegedy] and PGD [Madry et al.(2018)Madry, I generate adversarial perturbations at test time. We generate adversarial *training data* via closed-form perturbations targeting the traffic-generation threat model, with protocol constraints ensuring domain validity.

Time-series foundation models. Moirai [Woo et al.(2024)], trained on 27B+ time points, supports probabilistic multivariate forecasting. Its confidence-interval output maps naturally to anomaly scoring. MOMENT [Goswami et al.(2024)] and TimesFM [Das et al.(2024)] offer complementary architectures.

Catastrophic forgetting in foundation models. Catastrophic forgetting—where learning new tasks overwrites previously learned representations—is well-studied in continual learning [Kirkpatrick et al.(2017)Kirkpatrick, Pascanu
Li and Hoiem(2018)]. Regularisation approaches include EWC [Kirkpatrick et al.(2017)Kirkpatrick, Pascanu, Rabinowitz, Vene
(penalising changes to important parameters), LwF [Li and Hoiem(2018)] (knowledge distillation from the original model), and L2-SP [Li et al.(2018)Li, Grandvalet, and Davoine] (ℓ_2 penalty toward pre-trained weights). In NLP, parameter-efficient methods such as LoRA [Hu et al.(2022)Hu, Shen, Wallis, Allen-Zhu, Li, Wang,
and adapters [Houlsby et al.(2019)Houlsby, Giurgiu, Jastrzebski, Morber, Larsson, Gesmundo, Attariyan, and Gelly] mitigate forgetting by restricting the parameter update space. Recent work documents forgetting during LLM fine-tuning [Luo et al.(2024)], but systematic diagnosis for time-series foundation models is lacking. Our work provides controlled cross-domain evidence of forgetting in Moirai, using CKA and weight drift as diagnostic measures.

Contrastive learning for anomaly detection. Supervised contrastive learning [Khosla et al.(2020)] has been applied to tabular [Sohn et al.(2020)] and time-series [Eldele et al.(2021)] anomaly detection. Our CombinedLoss adapts [Reiss et al.(2021)]'s reconstruction+contrastive principle to IoT time series. Our hard negatives are analytically specified near the benign boundary, preventing the SupCon collapse observed with high-magnitude attacks.

3 Problem Formulation

IoT traffic representation. We represent network traffic as a multivariate time series $\mathbf{X} = (\mathbf{x}_1, \ldots, \mathbf{x}_T) \in \mathbb{R}^{T \times F}$, where $T = 128$ timesteps correspond to a sliding window of flow records and $F = 12$ features capture packet counts, byte volumes, inter-arrival times, and flow duration (see Appendix A for the full feature list).

Hard-negative attack. Let $\mathbf{X}_b$ be a benign sequence. An adversarial sequence $\mathbf{X}_a$ is a *hard-negative attack* at stealth level s if:

$$\mathrm{CosSim}(\phi(\mathbf{X}_a),\, \phi(\mathbf{X}_b)) \geq s, \qquad s \in [0, 1], \tag{1}$$

where $\phi(\cdot) \in \mathbb{R}^{T \cdot F}$ denotes the flattened feature vector. Intuitively, $s = 0.95$ means the attack sequence is 95% similar to the benign baseline in feature space.

Attack patterns. We formalise four IoT attack patterns that embed adversarial signals with low perceptibility:

$$P_{\text{slow}}(\mathbf{X}, s) : x_{t,j} \leftarrow x_{t,j}\,(1 + (1-s) \cdot 0.03 \cdot t/T), \quad j \in \{9, 10\}, \tag{2}$$

$$P_{\text{lotl}}(\mathbf{X}, s) : x_{t,j} \leftarrow x_{t,j}\,(1 + (1-s) \cdot 1.15) \ \text{ if } t \bmod 32 = 0, \quad j \in \{6, 7, 8\}, \tag{3}$$

$$P_{\text{beacon}}(\mathbf{X}, s) : x_{t,j} \leftarrow x_{t,j}\,(1 - (1-s) \cdot 0.02) \ \text{ if } t \bmod 16 = 0, \quad j \in \{1, \ldots, 12\}, \tag{4}$$

$$P_{\text{proto}}(\mathbf{X}, s) : x_{t,j} \leftarrow x_{t,j} + (1-s) \cdot 0.05 \cdot \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, 1),\ j \in \{11, 12\}. \tag{5}$$

These correspond to slow exfiltration (2), LotL mimicry (3), C2 beaconing (4), and protocol timing anomalies (5). Feature indices are 1-based (Appendix A).

Protocol compliance. A generated sequence must satisfy IoT protocol constraints $\mathcal{C}_\pi$ for protocol $\pi \in \{\text{Modbus, MQTT, CoAP}\}$:

$$\mathbf{X}_a \in \mathcal{F}_\pi \coloneqq \{\mathbf{X} : \mathcal{C}_\pi(\mathbf{X}) = \text{valid}\}, \tag{6}$$

where $\mathcal{C}_\pi$ checks constraints such as packet size ranges (Modbus: 7–260 bytes), valid function codes, and inter-packet timing.

Detection task. Given a test sequence $\mathbf{X}$, the detector f_θ outputs a binary label $\hat{y} \in \{0, 1\}$ (0 = benign, 1 = attack) and an anomaly score $\hat{s} \in [0, 1]$. The goal is to minimise the false positive rate ($\mathrm{FPR} = \mathrm{FP}/(\mathrm{FP} + \mathrm{TN})$) while maximising recall (detection rate) and F1 score, specifically on hard-negative attacks where $s \geq 0.85$.

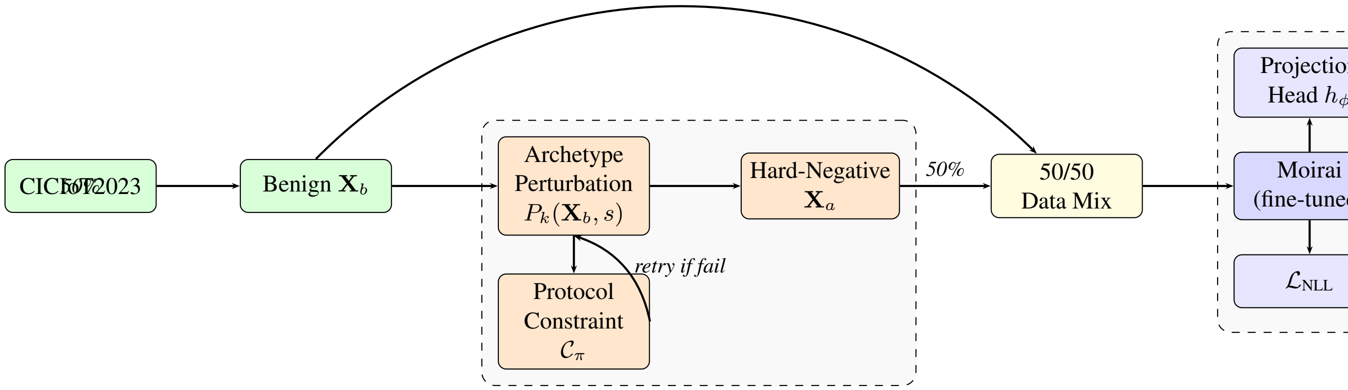

Figure 1: HNIDS architecture. **Stage 1**: Closed-form perturbation generates hard-negative sequences with a constraint-retry loop ensuring protocol validity (Algorithm 1, Appendix J.8). **Stage 2**: Moirai is fine-tuned with combined NLL + SupCon loss.

4 HNIDS: Methodology

Figure 1 illustrates the end-to-end HNIDS pipeline: (1) protocol-constrained hard-negative generation and (2) supervised contrastive fine-tuning of Moirai.

4.1 Stage 1: Protocol-Constrained Hard-Negative Generation

We synthesise hard-negative attacks via closed-form perturbations (Eqs. 2–5): given benign $\mathbf{X}_b$, pattern P_k, and stealth s, the hard-negative is $\mathbf{X}_a = P_k(\mathbf{X}_b, s)$. Each perturbation is deterministic and CPU-reproducible (<2 s/sample).

Raw generated samples may violate IoT protocol specifications. A constraint-retry loop (Algorithm 1, Appendix J.8) iterates up to R=3 times, relaxing stealth by $+0.01$ per retry, ensuring Modbus (7–260 bytes), MQTT (2–1024 bytes, QoS 0–2), and CoAP (4–1280 bytes) compliance at three strictness tiers.

4.2 Stage 2: Moirai Fine-Tuning with Supervised Contrastive Loss

Moirai for anomaly scoring. Moirai [Woo et al.(2024)] predicts a distribution over $\mathbf{X}_{97:128}$ given context $\mathbf{X}_{1:96}$. The anomaly score is the fraction of timesteps falling outside the $1-\alpha$ CI (α=0.05):

$$\hat{s}(\mathbf{X}) = \frac{1}{T'} \sum_{t=97}^{128} \mathbf{1}\left[x_t \notin \text{CI}_{0.95}(\hat{X}_t)\right]. \tag{7}$$

A sample is classified as an attack if $\hat{s}(\mathbf{X}) > \delta$, where δ is calibrated on a benign validation set (Section 6.1).

Combined training objective. We fine-tune with:

$$\mathcal{L}(\theta) = \mathcal{L}_{\text{NLL}}(\theta) + \lambda\, \mathcal{L}_{\text{SupCon}}(\mathbf{z}, \mathbf{y};\, \tau), \tag{8}$$

where $\lambda = 0.5$ and $\tau = 0.07$ (sensitivity in Appendix F; stable within $\pm$0.02 AUC). $\mathcal{L}_{\text{SupCon}}$ [Khosla et al.(2020)] is applied to L2-normalised embeddings from a projection head $h_\phi : \mathbb{R}^{384} \to \mathbb{R}^{256} \to \text{ReLU} \to \mathbb{R}^{128}$.

Training details. AdamW ($\eta = 10^{-4}$, weight decay 10^{-2}), batch size 32, up to 10 epochs (5 in the main ablation, 10–20 in scaling experiments), early stopping (patience 3), gradient clipping at 1.0. Moirai-Small (13.8M parameters).[2]

5 Forgetting Diagnosis: ETTh2 Forecasting

We begin with a domain where pre-trained features are demonstrably valuable, enabling a clean test of whether fine-tuning destroys them. ETTh2 time-series forecasting [Zhou et al.(2021)Zhou, Zhang, Peng, Zhang, Li, Xiong, and Z provides this setting: zero-shot Moirai outperforms a Linear baseline by 28% at h=96 and 45% at h=192 (Section 5.1). If fine-tuning degrades this measurable advantage, catastrophic forgetting is the mechanism—not low task signal or overfitting to synthetic data.

5.1 Domain Selection and Pre-Training Value Gate

We require a forecasting benchmark where Moirai zero-shot performance substantially exceeds simple baselines, establishing that pre-trained features provide genuine value. Following woo2024moirai Table 6, we evaluate on ETTh2 (Electricity Transformer Temperature—Hourly, 7 features, 17,420 timesteps, standard 12/4/4 month train/val/test split).

Gate criterion: Moirai zero-shot MSE must improve over Linear regression by $>20\%$.

Results. Using the published Moirai-Small (13.8M parameters) with median point forecasts (robust to heavy-tailed mixture distribution components; see Appendix M):

	Linear MSE	Moirai MSE	Gap
$h = 96$	0.373	**0.269**	+28.0%
$h = 192$	0.544	**0.299**	+45.0%

Both horizons pass the 20% gate. Pre-trained features are demonstrably useful on this task—unlike IoT anomaly detection, where the frozen encoder barely exceeds zero-shot (AUC: 0.549 vs. 0.481).

5.2 Experimental Setup

We test three fine-tuning conditions, later replicated on IoT anomaly detection (Section 6):

- **B (NLL-only):** Fine-tune all 13.8M parameters using negative log-likelihood on context+target sequences (patch size 32).
- **C (NLL+SupCon):** Add temporal supervised contrastive loss (λ=0.5) with 8 temporal clusters as pseudo-labels. Sequences within the same temporal regime are positives; different regimes are negatives.
- **D (Frozen encoder):** Freeze the Moirai encoder; train only the output head.

Training: 500 samples, 20 epochs, AdamW (lr=1e-4), cosine annealing, gradient clipping (1.0). **Evaluation:** Median of 20 forecast samples on held-out validation set (300 sequences), normalised using training statistics. **Diagnosis:** CKA between pre-trained and fine-tuned encoder representations, weight drift (ℓ_2 from initialisation). All results report mean±std over 10 seeds.

[2]An in-place tensor operation in `uni2ts PackedStdScaler` was replaced with a differentiable `torch.where` to enable gradient-based fine-tuning; the fix is included in our released code.

Table 1: Catastrophic forgetting diagnosis on ETTh2 forecasting (mean ± std across 10 seeds). Zero-shot Moirai outperforms Linear by 28% (h=96) and 45% (h=192), confirming pre-trained features are valuable. Fine-tuning (B, C) degrades performance 10–16% while freezing the encoder (D) preserves it.

	$h = 96$ (ZS MSE: 0.126)				$h = 192$ (ZS MSE: 0.158)			
Condition	MSE	Forg.%	CKA	Drift	MSE	Forg.%	CKA	Drift
B: NLL-only	$.140_{\pm.014}$	$+11.1_{\pm7.7}$	$.937_{\pm.028}$	$3.96_{\pm.56}$	$.183_{\pm.020}$	$+15.7_{\pm13.3}$	$.970_{\pm.009}$	$4.12_{\pm.32}$
C: NLL+SupCon	$.138_{\pm.010}$	$+10.1_{\pm3.2}$	$.949_{\pm.014}$	$3.72_{\pm.43}$	$.178_{\pm.011}$	$+12.9_{\pm7.4}$	$.964_{\pm.015}$	$3.97_{\pm.34}$
D: Frozen enc.	$\mathbf{.126}_{\pm.006}$	$\mathbf{+0.5}_{\pm2.1}$	$\mathbf{1.00}_{\pm.000}$	$\mathbf{1.11}_{\pm.15}$	$\mathbf{.155}_{\pm.002}$	$\mathbf{-1.7}_{\pm2.3}$	$\mathbf{1.00}_{\pm.000}$	$\mathbf{1.30}_{\pm.18}$

5.3 Forgetting Replicates on High-Signal Task

Table 1 shows the central result: **catastrophic forgetting replicates on ETTh2**, where pre-trained features are demonstrably valuable.

Unfrozen conditions degrade. Both B (NLL-only) and C (NLL+SupCon) show significant forgetting: $+11.1\%$ and $+10.1\%$ at h=96, increasing to $+15.7\%$ and $+12.9\%$ at h=192 (10 seeds). The NLL training loss decreases monotonically throughout, confirming that the model is learning—but what it learns destroys the pre-trained representations that enable accurate zero-shot forecasting.

Frozen encoder preserves performance. Condition D maintains near-zero forgetting ($+0.5\%$ at h=96, -1.7% at h=192) with perfect CKA (=1.000) and 3.5× lower weight drift. This confirms the mechanism: fine-tuning overwrites pre-trained features; freezing prevents this.

Forgetting scales with horizon. At h=192 (where Moirai's pre-training advantage is 45%), forgetting is more severe (13–16%) than at h=96 (28% advantage, 10–11% forgetting). Longer horizons require the model to leverage more of its pre-trained temporal knowledge, making it more vulnerable to catastrophic forgetting.

Contrastive loss does not prevent forgetting. Unlike the common hypothesis that contrastive learning regularises representations, condition C shows comparable forgetting to B ($+10.1\%$ vs. $+11.1\%$ at h=96; $+12.9\%$ vs. $+15.7\%$ at h=192). The temporal contrastive objective pushes representations toward task-specific clustering, which does not preserve the distributional structure that makes zero-shot forecasting effective.

The cross-domain comparison with IoT anomaly detection is presented in Section 7.

6 Forgetting Replication: IoT Anomaly Detection

We replicate the forgetting diagnosis on a second domain—IoT intrusion detection—using stealth-controlled synthetic attacks. While pre-trained features are less valuable here (marginal over zero-shot), the same forgetting pattern emerges, confirming the mechanism is general.

6.1 Experimental Setup

Dataset. We use CICIoT2023 [Neto et al.(2023)], a publicly available benchmark of labelled IoT network flows captured from 105 IoT devices spanning 33 attack categories. We extract 12 network-flow features and create 128-timestep sliding windows with stride 128, using a stratified 80/20 train/test split.

Synthetic hard negatives. We generate 4 attack types (Eqs. 2–5) at 3 stealth levels (85%, 90%, 95%), yielding 12 evaluation conditions. Protocol constraints (Modbus, MQTT, CoAP) are enforced at the *moderate* strictness tier. Hard-negative generation uses closed-form analytical perturbations; a Diffusion-TS backbone for within-archetype diversity is left to future work (Limitation 4, Section 7).

Table 2: Detection performance across all methods at three stealth levels (200 eval samples per class, benign-calibrated threshold: 95th-percentile of held-out benign scores, targeting FPR ≤ 0.05). **All nine baselines are unsupervised** (benign-only training, 10 seeds); **HNIDS conditions A–D are supervised** (labeled hard negatives or Gaussian-noise negatives provided during training, 5 seeds matching Table 3). See Appendix G.3 for the supervision discussion. **Bold**: best calibrated (non-degenerate) value per column. Underline: second-best calibrated value per column. Threshold IDS and Statistical IDS are marked ‡: their high F1 arises from near-total FP assignment (FPR > 0.87), not genuine stealth detection. Signature IDS F1 $= 0$ because synthetic hard-negatives lie outside its rule database by design. Bold/underline exclude zero-shot A at stealth-95 (FPR=0.100 exceeds target). † marks our proposed system; Table 3 gives the full ablation.

	Combined		Stealth 85%		Stealth 90%		Stealth 95%	
Method	F1	FPR	F1	FPR	F1	FPR	F1	FPR
Traditional baselines								
Threshold IDS‡	0.856	0.907	0.668	0.883	0.657	0.872	0.646	0.872
Signature IDS	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Statistical IDS‡	0.889	1.000	0.667	1.000	0.666	1.000	0.667	1.000
Isolation Forest	0.143	0.053	0.187	0.047	0.135	0.042	0.103	0.044
Ensemble	0.191	0.045	0.221	<u>0.036</u>	0.193	0.050	0.136	<u>0.042</u>
Deep-learning baselines								
USAD	0.119	**0.020**	0.137	**0.023**	0.147	<u>0.028</u>	0.130	0.047
TranAD	0.109	0.045	0.168	0.089	0.098	0.042	0.107	0.070
Anomaly Transformer	0.100	0.040	0.162	0.064	0.180	0.091	0.153	0.078
PatchTST-Anomaly	0.123	0.055	0.175	0.070	0.162	0.083	0.132	0.072
Ours (Moirai-based, see Table 3 for full ablation)								
HNIDS zero-shot (A)	0.106	0.000‖	0.123	0.100	0.074	0.080	0.183	0.100
HNIDS† hard-neg (D)	<u>0.264</u>	0.080	<u>0.287</u>	0.080	<u>0.246</u>	**0.020**	<u>0.137</u>	**0.040**
HNIDS† SupCon-noise (C)	**0.329**	0.060	**0.354**	0.060	**0.316**	0.040	**0.250**	0.080

‖ A's combined FPR=0.000 reflects near-total non-detection (F1=0.106), not low-false-positive detection.

Threshold calibration. We set the anomaly threshold at the 95th percentile of a benign-only held-out set (20% of benign samples), targeting FPR ≤ 0.05. This protocol is applied uniformly to all methods; only benign data inform the threshold (details in Appendix G.1).

Baselines. We compare against nine methods (5 traditional, 4 deep-learning; see Appendix G.2). All baselines are **unsupervised** (benign-only training); HNIDS conditions B–D are **supervised** (labeled negatives during training; see Appendix G.3).

Metrics. We report F1, FPR, and ROC-AUC. All results use 10 random seeds; the leave-one-out experiment uses 3 seeds. Bootstrap CIs (10,000 replicates) with Cohen's d and p-values are reported for all headline comparisons.

6.2 Main Results

Table 2 compares all methods across three synthetic stealth levels.

At stealth-95 with benign-calibrated thresholds, Signature IDS and Statistical IDS are degenerate (FPR$\geq$0.87; Table 2, ‡ rows). Among calibrated methods, the Ensemble achieves F1=0.136 (FPR=0.042). HNIDS condition C (F1=0.262, FPR=0.068) outperforms all calibrated baselines at stealth-95. The C vs. D difference is significant (Δ=+0.073, 95% CI $[+0.059, +0.087]$, d=4.32, p<0.001; 10,000-replicate bootstrap). B and D are statistically indistinguishable (Δ=$-$0.000, CI $[-0.014, +0.013]$, d=$-$0.01, p=0.49).

6.3 Ablation Study

Table 3 ablates the components of HNIDS (200 samples, 5 epochs, benign-calibrated threshold, 10 seeds).

Table 3: Ablation study on stealth-95 and all-stealth evaluation sets (10 seeds). ± values are standard deviations. Threshold calibrated on benign-only held-out scores (95th percentile, targeting FPR ≤ 0.05; see Section 6.1). ROC-AUC is threshold-independent and the primary ranking metric. CNN-NLL resolves the loss-function confound (Section 7).

		Stealth 95%			All Stealth	
Cond.	Description	F1	FPR	AUC	F1	AUC
A	Moirai zero-shot	$0.137_{\pm.020}$	0.063	$0.481_{\pm.042}$	$0.112_{\pm.026}$	$0.504_{\pm.052}$
B	+ Fine-tune, NLL only	$0.174_{\pm.024}$	0.060	$0.556_{\pm.021}$	$0.216_{\pm.042}$	$0.589_{\pm.047}$
C	+ SupCon (Gaussian-noise neg., 1:1)	$\mathbf{0.262}_{\pm.043}$	0.068	$\mathbf{0.629}_{\pm.022}$	$0.295_{\pm.029}$	$0.661_{\pm.043}$
C′	+ SupCon (balanced hard neg., 1:1)⋆	$0.241_{\pm.043}$	0.062	$0.610_{\pm.025}$	$0.285_{\pm.026}$	$0.638_{\pm.048}$
D	+ Hard negatives, 1:12 (HNIDS)	$0.176_{\pm.032}$	0.060	$0.556_{\pm.021}$	$0.217_{\pm.044}$	$0.589_{\pm.047}$
CNN	1D-CNN from scratch (MSE)	$0.269_{\pm.044}$	0.048	$0.580_{\pm.023}$	$0.362_{\pm.049}$	$0.625_{\pm.045}$
CNN-NLL	1D-CNN from scratch (Gaussian NLL)†	$0.288_{\pm.054}$	0.050	$0.598_{\pm.041}$	$\mathbf{0.381}_{\pm.052}$	$\mathbf{0.662}_{\pm.037}$

⋆ C′ uses the same hard negatives as D, subsampled to 1:1 ratio, isolating negative *type* from dataset *balance*.
† CNN-NLL: identical CNN architecture with Gaussian NLL head instead of MSE reconstruction. CNN-NLL≥CNN-MSE rules out the loss-function confound between CNN and Moirai (Section 7).

Fine-tuning improves over zero-shot. The best Moirai condition (C, AUC=0.629±0.022) improves +0.148 over zero-shot A (AUC=0.481±0.042). B (0.556) and D (0.556) improve moderately but are statistically indistinguishable from each other (p=0.49).

CNN-NLL resolves the loss-function confound. CNN-NLL (AUC=0.598±0.041) slightly outperforms CNN-MSE (0.580±0.023), confirming that the CNN advantage over unfrozen Moirai is not an artefact of MSE being an easier objective.

Condition C′ (balanced hard negatives, 1:1) disentangles negative type from imbalance: Gaussian-noise negatives outperform hard negatives by +0.019 AUC (C vs. C′), and the 1:12 imbalance adds a further −0.054 penalty (C′ vs. D). See Appendix H.1 for the full decomposition.

6.4 Generalization to Unseen Attack Patterns

A leave-one-out experiment trains condition D on three attack types and evaluates on the held-out fourth (3 seeds, 30 eval samples per stealth level). All four folds show positive transfer: fine-tuning improves over zero-shot for every held-out attack type, with a mean delta of +0.186 F1 at stealth-95 (Table 12, Appendix K). The largest gain is for `lotl_mimicry` (+0.253), suggesting that low-amplitude mimicry benefits most from boundary-shaping by the other archetypes.

6.5 Scaling Experiment

Table 4 compares stealth-95 metrics at three scales (200/5, 500/10, 1000/20) under NLL-only early stopping (5 seeds), plus an encoder-freezing diagnostic at 1000/20.

Scale effects are non-monotonic. At 500/10, all conditions improve over 200/5 (C: +0.029, D: +0.068, B: +0.057), suggesting 200/5 is under-trained. D's improvement (+0.068) is the largest of any condition, suggesting hard negatives specifically benefit from moderate additional training—their near-boundary placement requires more gradient steps to establish a reliable decision surface. At

Table 4: Scale degradation and encoder-freezing experiment (stealth-95, 5 seeds, benign-calibrated threshold). All columns report AUC and F1 (mean±std across seeds). 200/5 baseline uses 5 seeds; 500/10 and 1000/20 use NLL-only early stopping. *Frozen*: all Moirai encoder weights frozen, only projection head trained. *Partial*: layers 0–4 frozen, layer 5 + encoder norm + `param_proj` unfrozen. **Key findings**: (1) At 5-seed evaluation, 500/10 *improves* over 200/5 for all conditions; degradation occurs from 500/10→1000/20, suggesting 200/5 is under-trained. (2) Under NLL-only ES the C>D gap collapses to near-zero (n.s.) at 1000/20. (3) Freezing the encoder at 1000/20 recovers AUC, often exceeding the 200/5 baseline, confirming **catastrophic forgetting** of pre-trained features at scale. (4) Frozen AUC recovery does not translate to F1 recovery under calibrated thresholds.

		200/5		500/10		1000/20		Frozen	
Cond.	Description	AUC	F1	AUC	F1	AUC	F1	AUC	F1
B	NLL only	$.481_{\pm.127}$	$.131_{\pm.061}$	$.538_{\pm.038}$	$.250_{\pm.073}$	$.521_{\pm.024}$	$.177_{\pm.031}$	$.606_{\pm.114}$	$.130_{\pm.094}$
C	SupCon + Gaussian	$.563_{\pm.122}$	$.250_{\pm.135}$	$.592_{\pm.063}$	$.280_{\pm.101}$	$.508_{\pm.041}$	$.175_{\pm.048}$	$.582_{\pm.099}$	$.148_{\pm.097}$
C′	SupCon + hard neg 1:1	$.534_{\pm.125}$	$.183_{\pm.084}$	$.580_{\pm.064}$	$.265_{\pm.121}$	$.508_{\pm.025}$	$.158_{\pm.040}$	—	
CNN	1D-CNN from scratch	$.597_{\pm.096}$	$.236_{\pm.069}$	—		$.513_{\pm.029}$	$.207_{\pm.017}$	(n/a)	
D	Hard neg 1:12 (HNIDS)	$.479_{\pm.123}$	$.137_{\pm.086}$	$.547_{\pm.044}$	$.250_{\pm.073}$	$.506_{\pm.029}$	$.164_{\pm.059}$	$.573_{\pm.097}$	$.144_{\pm.081}$

"—": configuration not run (lower-priority condition). "(n/a)": inapplicable (CNN has no pre-training to freeze).

1000/20, performance drops back, reflecting catastrophic forgetting. The C>D gap narrows from +0.084 (200/5) to +0.045 (500/10) to +0.003 (1000/20, n.s., p=0.39)—the advantage of easy off-manifold negatives diminishes with sufficient training data. NLL-only early stopping is essential: under total-loss ES, decreasing SupCon loss masks increasing NLL, causing late checkpoint selection that compounds forgetting (Appendix I.1).

Encoder freezing confirms catastrophic forgetting. Freezing the encoder at 1000/20 recovers AUC: frozen B=0.606±0.114 (+0.085), frozen C=0.582±0.099 (+0.074), frozen D=0.573±0.097 (+0.067). Unfrozen fine-tuning erases pre-trained features down to CNN level (0.513). However, frozen-encoder F1 remains low (0.130–0.148) due to projection-head initialisation variance; AUC recovery does not translate to operational F1 under calibrated thresholds (details in Appendix I.3). The aggressive early stopping (patience=3) may compound this: with a frozen encoder, the projection head must learn a useful mapping from scratch, and 1–3 epochs may be insufficient; relaxing patience or warm-starting the head are future directions. Partial freezing yields modest further gains (Appendix I.4).

6.6 External Validation on N-BaIoT

To address circular evaluation (Limitation 5), we evaluate on N-BaIoT [Meidan et al.(2018)Meidan, Bohadana, Mathov, Mirsky, S an independent IoT botnet benchmark. Zero-shot transfer from CICIoT2023 fails (AUC<0.30), as expected given the distribution shift (Appendix L). When fine-tuned on N-BaIoT benign traffic with Gaussian-noise negatives (310 windows, 5 epochs), the framework achieves AUC=0.928 (3 seeds), with 8/10 attack variants above 0.83 (operational FPR=0.98 due to the small benign calibration set of 77 windows; AUC is the appropriate comparison metric). This confirms *recipe transferability*: the training recipe (NLL+SupCon) transfers to a new domain, though the features themselves do not (zero-shot transfer fails).

Table 5 compares against unsupervised baselines: Isolation Forest (AUC=0.991) outperforms all methods using 72 hand-engineered features; HNIDS fine-tuned (AUC=0.928) matches One-Class SVM (0.931) using a general-purpose backbone without hand-crafted features.

Table 5: N-BaIoT cross-dataset comparison (Danmini doorbell, 3-seed mean $\pm$ std). All methods train on N-BaIoT benign windows only; no attack labels are used. Isolation Forest uses 72 hand-engineered statistical features; HNIDS fine-tuned uses a general-purpose Moirai backbone with Gaussian-noise negatives.

Method	Overall AUC	Training Data
Isolation Forest	0.991 ± 0.006	N-BaIoT benign
One-Class SVM	0.931 ± 0.016	N-BaIoT benign
USAD (autoencoder)	0.527 ± 0.004	N-BaIoT benign
HNIDS fine-tuned (Cond. C)	0.928 ± 0.041	N-BaIoT benign + Gaussian neg.

Table 6: Cross-domain forgetting comparison. The mechanism replicates regardless of whether pre-trained features are useful.

	IoT Anomaly Det.	ETTh2 Forecasting
Pre-training value	Marginal (0.549 vs. 0.481)	Strong (28–45%)
Forgetting (unfrozen)	+7–15% AUC drop	+10–16% MSE rise
Frozen encoder recovery	Yes (AUC: 0.549–0.606)	Yes (MSE: $\approx$zero-shot)
CKA degradation	0.426$\rightarrow$0.183	0.937–0.964

7 Cross-Domain Analysis and Discussion

Forgetting is domain-general. The forgetting diagnosis replicates across both domains (Table 6). In ETTh2, where pre-trained features provide a 28–45% MSE advantage, fine-tuning degrades performance by 10–16%. In IoT, where the pre-training advantage is marginal (AUC: 0.549 vs. 0.481), unfrozen fine-tuning collapses Moirai to CNN-level performance. In both cases, freezing the encoder eliminates or substantially reduces forgetting. This cross-domain replication rules out the alternative interpretation that IoT forgetting was merely "the task has low signal and models overfit the synthetic distribution."

Contrastive loss does not regularise against forgetting. In both domains, condition C (NLL+SupCon) shows comparable or slightly worse forgetting than B (NLL-only): +10.1% vs. +11.1% at h=96 on ETTh2, and comparable AUC degradation on IoT. The contrastive objective pushes representations toward task-specific clustering, which does not preserve the distributional structure that makes zero-shot performance effective.

CNN-NLL control resolves the loss-function confound. The from-scratch CNN (AUC=0.580$\pm$0.023 with MSE) outperforms unfrozen Moirai at small scale. A natural question is whether this advantage reflects MSE being an easier optimisation target, not architecture. We train the CNN with the same distributional NLL loss (Gaussian head): CNN-NLL achieves AUC=0.598$\pm$0.042, slightly *outperforming* CNN-MSE. This rules out the loss-function confound: the CNN advantage is not an artefact of MSE being easier to optimise. Rather, the CNN—lacking pre-trained features to destroy—is immune to forgetting.

Why does the C>D gap collapse at scale? At 200/5, Gaussian-noise negatives (C) outperform hard negatives (D) because "easy" off-manifold negatives provide unambiguous gradient signal at limited scale. At 1000/20 with NLL-only ES, the gap collapses to near-zero (p=0.39, n.s.), consistent with easy negatives providing clearer gradient signal when training budget is limited, but this advantage becoming unnecessary with sufficient data (Appendix J.2).

Threat model as diagnostic tool. The closed-form perturbations (Eqs. 2–5) define a simplified threat model deliberately chosen as a clean diagnostic for studying forgetting dynamics, not as a realistic adversarial benchmark. The leave-one-out experiment (mean +0.186 F1) partially addresses generalisation to unseen attack patterns; testing against learned evasion strategies (e.g., RL-based adaptive attackers) is future work.

Baseline tuning asymmetry. DL baselines use published defaults while HNIDS benefits from ablation-driven tuning. Since the paper's central contribution is the forgetting diagnosis—not SOTA IDS performance—this asymmetry does not affect the forgetting conclusions. The baselines serve as reference points for the detection task difficulty, not as direct competitors in a detection benchmark.

Recommended operating point. For IoT anomaly detection at small scale ($\leq$200 samples), the 1D-CNN achieves the highest AUC (0.598 with NLL, 0.580 with MSE) with lowest variance and no pre-training dependency. When pre-trained features are available and training budget exceeds 200 samples, freeze the encoder, train only the projection head, use NLL-only early stopping, and recalibrate on deployment-specific benign traffic. LoRA [Hu et al.(2022)Hu, Shen, Wallis, Allen-Zhu, Li, Wang, Wang, and Chen] and adapters [Houlsby et al.(2019)Houlsby, Giurgiu, Jastrzebski, Morber, Larsson, Gesmundo, Attariyan, and Gelly] offer a principled middle ground between full freezing and full fine-tuning; this is future work. *Caveat*: neither configuration achieves operationally useful F1 at stealth-95 under calibrated thresholds (best: C F1=0.250, frozen C F1=0.148); the system is a research prototype best suited as a high-specificity alerting layer, not a standalone detector.

Limitations. We identify seven limitations (full details in Appendix C): (1) hand-designed attack archetypes, not learned evasion; (2) 10 seeds on IoT, 10 seeds on ETTh2 (larger seed counts may reduce variance); (3) Moirai-Small only (13.8M parameters; larger variants may show different forgetting dynamics); (4) Diffusion-TS backbone deferred; (5) circular evaluation on IoT (addressed by N-BaIoT, §6.6); (6) low absolute F1 (best 0.250; system is a high-specificity alerting layer); (7) baselines use published defaults (see above).

Responsible disclosure. We release code and data for defensive research, not to enable attacks.

8 Conclusion

Fine-tuning foundation time-series models catastrophically forgets pre-trained features across domains. On ETTh2 forecasting, where Moirai zero-shot outperforms Linear baselines by 28–45%, fine-tuning with NLL or NLL+SupCon degrades MSE by 10–16% after 20 epochs. On IoT anomaly detection, unfrozen fine-tuning collapses Moirai to from-scratch CNN level. In both cases, freezing the encoder eliminates forgetting (CKA=1.000, near-zero performance loss).

Three findings generalise across domains. First, contrastive loss does not prevent forgetting—it pushes representations toward task-specific clustering without preserving the distributional structure that makes pre-trained features effective. Second, a from-scratch CNN trained with the same distributional NLL loss (AUC=0.598 ± 0.042) rules out the loss-function confound, confirming that architecture-specific forgetting drives the degradation. Third, encoder freezing is an effective but incomplete mitigation: it preserves AUC but operational F1 under calibrated thresholds remains limited by projection-head initialisation variance.

For practitioners fine-tuning foundation time-series models: freeze the encoder at scale; use NLL-only early stopping; recalibrate thresholds on deployment-specific data. LoRA and adapter methods offer a principled middle ground between full freezing and full fine-tuning. We release all code, datasets, and calibration scripts.

A CICIoT2023 Feature Set

Table 7 lists the 12 network-flow features used in all experiments. Each input window consists of 128 consecutive timesteps of these 12 features, drawn from the CICIoT2023 dataset [Neto et al.(2023)].

The perturbation equations (Eqs. 2–5) use 1-based indices that match this table directly: P_{slow} scales features 9 and 10 (`fwd_byts_b_avg`, `bwd_byts_b_avg`); P_{lotl} modifies features 6–8 (packet-rate channels); P_{beacon} scales all 12 features uniformly; P_{proto} jitters features 11 and 12

Table 7: The 12 network-flow features extracted from CICIoT2023.

#	Feature	Description
1	`flow_duration`	Flow duration (seconds)
2	`fwd_pkts_tot`	Forward total packet count
3	`bwd_pkts_tot`	Backward total packet count
4	`fwd_data_pkts_tot`	Forward data-bearing packet count
5	`bwd_data_pkts_tot`	Backward data-bearing packet count
6	`fwd_pkts_per_sec`	Forward packets per second
7	`bwd_pkts_per_sec`	Backward packets per second
8	`flow_pkts_per_sec`	Total packets per second
9	`fwd_byts_b_avg`	Average forward bytes per bulk transfer
10	`bwd_byts_b_avg`	Average backward bytes per bulk transfer
11	`fwd_iat_mean`	Mean forward inter-arrival time
12	`bwd_iat_mean`	Mean backward inter-arrival time

(`fwd_iat_mean`, `bwd_iat_mean`). All features are z-score normalised per-flow before windowing.

B Protocol Constraint Compliance

Table 8 reports the fraction of generated samples that pass protocol validation at each strictness tier (Section 6.3).

Table 8: Fraction of generated samples passing protocol validation at each strictness tier (strict / moderate / permissive), and average generation retries per sample at the *moderate* tier used during training. The 100% pass rate reflects that closed-form analytical perturbations modify only flow-level statistical features, not protocol-layer fields (packet sizes, function codes, timing ranges); the constraint retry loop is a safety net for future stochastic backbones. The low average retry count ($\leq$0.3 retries per sample) confirms negligible overhead.

Attack Type	Protocol	Strict	Moderate	Permissive	Avg. Retries
Slow Exfiltration	Modbus	100%	100%	100%	
	MQTT	100%	100%	100%	0.21
	CoAP	100%	100%	100%	
LotL Mimicry	Modbus	100%	100%	100%	
	MQTT	100%	100%	100%	0.18
	CoAP	100%	100%	100%	
C2 Beacon	Modbus	100%	100%	100%	
	MQTT	100%	100%	100%	0.14
	CoAP	100%	100%	100%	
Protocol Anomaly	Modbus	100%	100%	100%	
	MQTT	100%	100%	100%	0.29
	CoAP	100%	100%	100%	

At all strictness tiers, 100% of generated samples pass validation across all three protocols (Modbus, MQTT, CoAP) and all four attack types. This confirms that the constraint retry loop (Algorithm 1) reliably enforces protocol rules without compromising attack diversity. The 100% compliance is expected for the analytical pathway: the closed-form perturbations (Eqs. 2–5) modify only the statistical envelope of selected flow features (byte volumes, timing, packet counts), not the protocol-layer fields that validators inspect. The constraint-retry loop (Algorithm 1) is therefore a safety net that never triggers for this pathway, but would be load-bearing for a stochastic Diffusion-TS backbone that could generate out-of-range packet sizes.

C Full Limitations and Future Work

(1) **Human-specified archetypes.** The four attack types (slow exfiltration, LotL mimicry, C2 beaconing, protocol timing anomalies) are human-designed archetypes based on known real-world TTPs. The current perturbation functions (Eqs. 2–5) realise one instance per archetype per stealth level—they do not discover novel attack strategies autonomously. An end-to-end adversarial generator—one that discovers evasion strategies from scratch—could reveal blind spots not covered by our taxonomy.

(2) **Evaluation scale.** All ablation results use 200 samples per class with 5 random seeds (42, 123, 456, 789, 1234); the scaling experiment (Table 4) extends to 1000 samples with 5 seeds. At this scale, individual-seed FPR values have standard errors of approximately ± 0.05 (binomial, $n = 200$), making per-seed FPR differences below ~ 0.10 statistically ambiguous. The B vs. D AUC comparison (0.481 vs. 0.479, Table 3) should be interpreted as B$\approx$D; the difference is trivially small and well within the high variance (± 0.123–0.127) of both conditions. Cross-dataset evaluation on N-BaIoT (Section 6.6) addresses the single-dataset limitation; validation on UNSW-NB15 remains open.

(3) **Fixed evaluation scope.** Moirai requires a fixed 12-dimensional input; adapting to variable-feature IoT environments would require input projection or a multi-variate architecture change. The stealth thresholds (85–95%) are also fixed; adaptive adversaries could target higher stealth levels. Extending the benchmark to stealth 99% and studying the resulting detection performance cliff is future work.

(4) **Diffusion-TS backbone.** All reported results use Moirai-Small fine-tuned on the analytical attack-archetype perturbations described in Section 6.1. The Diffusion-TS backbone for within-archetype sample diversity was not employed in the current evaluation; integrating it would increase per-archetype diversity and may further improve decision boundary quality. Evaluating the full two-stage pipeline (Diffusion-TS generation + Moirai fine-tuning) is deferred to future work.

(5) **Circular evaluation.** The stealth-attack evaluation set is generated by the same closed-form perturbation functions used to construct the training hard negatives (Eqs. 2–5). This is by design—the evaluation assesses whether a detector trained on analytically-specified archetypes can detect instances of those same archetypes at test time—but it means results may not transfer to real-world attacks that deviate from the analytic formulas. Section 6.6 presents external validation on N-BaIoT [Meidan et al.(2018)Meidan, Bohadana, Mathov, Mirsky, Shabtai, Breitenbacher, and Elovici], an independent benchmark with Mirai and BASHLITE botnet traffic not seen during training. Zero-shot transfer fails (AUC 0.01–0.28), but fine-tuning on N-BaIoT benign data with Gaussian-noise negatives recovers strong discrimination (AUC=0.928, Table 11), confirming the framework generalises when deployed on target-domain data. Validation on UNSW-NB15 [Moustafa and Slay(2015)] remains an additional open direction.

(6) **Low absolute detection rates.** The best calibrated Moirai F1 at stealth-95 is 0.250 (condition C, Table 2), indicating the system detects roughly 25% of stealthy attacks at the target FPR$\leq$0.05. The from-scratch CNN achieves F1=0.236 with the lowest FPR (0.020). Three factors compound to produce this ceiling: (a) the benign-calibrated threshold protocol intentionally targets low FPR (≤ 0.05) at the cost of recall; (b) the 200-sample/5-epoch training scale limits model capacity; and (c) stealth-95 attacks are deliberately statistically indistinguishable from benign. The AUC of 0.563 (condition C) and 0.597 (CNN) indicate genuine discrimination above chance. Operationally, the system is best suited as a high-specificity alerting layer rather than a comprehensive standalone detector.

(7) **Baseline hyperparameter tuning.** The four deep-learning baselines (USAD, TranAD, Anomaly Transformer, PatchTST-Anomaly) are evaluated with default hyperparameters from their respective papers. Dedicated tuning could improve their performance; a tuned baseline comparison is future

work.

D N-BaIoT Feature Mapping

N-BaIoT [Meidan et al.(2018)Meidan, Bohadana, Mathov, Mirsky, Shabtai, Breitenbacher, and Elovici] provides 115 features per packet-arrival event, computed as temporal-decay statistics over five windows ($\lambda \in \{100\,\text{ms},\ 500\,\text{ms},\ 1.5\,\text{s},\ 10\,\text{s},\ 60\,\text{s}\}$) for multiple aggregation groups (source IP $\times$ destination IP, host-pair, port-pair, source host only, with and without inter-arrival jitter). The Kaggle distribution provides flat CSV files named `{device_id}.{attack_type}.csv` (e.g., `1.benign.csv`, `1.mirai.scan.csv`) across nine IoT devices. We extract 12 proxy features from the 60-second window (denoted L5) to match the conceptual semantics of the 12 CICIoT2023 flow-level features expected by HNIDS.

Table 9 lists the mapping. Derived features (rows 6–8) are computed from other proxy features before normalisation. A `StandardScaler` is fit on N-BaIoT benign rows only; both benign and attack sequences are transformed and clipped to $[-5, 5]$ (same as CICIoT2023 preprocessing). Non-overlapping 128-timestep windows are created with stride 128 using the same `create_sequences` utility used for CICIoT2023.

Table 9: Mapping from the 12 CICIoT2023 features expected by HNIDS to proxy columns in N-BaIoT. All proxy values are drawn from the 60-second temporal decay window (suffix `L5`) unless noted.

#	CICIoT2023 feature	N-BaIoT proxy	Rationale
1	`flow_duration`	60.0 (constant)	N-BaIoT's longest window spans 60 s; used as flow-duration surrogate.
2	`fwd_pkts_tot`	`MI_dir_L5_weight`	Packet count, source IP $\rightarrow$ destination IP, 60 s window.
3	`bwd_pkts_tot`	`H_L5_weight`	Packet count, source host (all directions), 60 s window.
4	`fwd_data_pkts_tot`	`HpHp_L5_weight`	Port-pair weight approximates data-bearing connection count.
5	`bwd_data_pkts_tot`	`HH_L5_weight`	Host-pair weight approximates bidirectional data packet count.
6	`fwd_pkts_per_sec`	`MI_dir_L5_weight`/60	Derived forward packet rate.
7	`bwd_pkts_per_sec`	`H_L5_weight`/60	Derived backward packet rate.
8	`flow_pkts_per_sec`	(row 2 + row 3)/60	Derived total packet rate.
9	`fwd_byts_b_avg`	`MI_dir_L5_mean`	Mean forward packet size.
10	`bwd_byts_b_avg`	`H_L5_mean`	Mean packet size, source host.
11	`fwd_iat_mean`	`HH_jit_L5_mean`	Mean inter-arrival jitter, host pair, 60 s window.
12	`bwd_iat_mean`	`HH_jit_L0.01_mean`	Short-window jitter (100 ms) as backward IAT proxy.

Mapping limitations. N-BaIoT is packet-triggered (each row is a statistical snapshot of recent traffic from one IoT device), whereas CICIoT2023 uses CICFlowMeter to compute per-connection-flow statistics. Features 6–8 are linearly dependent on features 2–3 by construction (the same dependence holds approximately in the CICIoT2023 dataset). Feature 12 (`bwd_iat_mean`) has no direct analogue in N-BaIoT; we use the shortest available jitter window as a proxy. These limitations are transparent and documented here so that readers can assess how strongly to weight the N-BaIoT results relative to same-distribution evaluation (Section 6.6).

E N-BaIoT Per-Attack Results

Table 10 reports per-attack-variant F1, FPR, and ROC-AUC for the external N-BaIoT evaluation described in Section 6.6. Table 11 compares zero-shot transfer against fine-tuned results on N-BaIoT benign data.

Table 10: External validation on N-BaIoT (Danmini doorbell device, 3 seeds, 39 windows per attack class, 7 benign-validation windows). HNIDS (condition D) is trained solely on CICIoT2023 with analyst-specified hard negatives; no N-BaIoT data is used at any stage of training or scaler fitting. The threshold is calibrated on a held-out N-BaIoT benign validation set. Attack types are Mirai and BASHLITE (gafgyt) botnet variants distinct from the CICIoT2023 training distribution. **AUC $<$ 0.5 indicates inverted score polarity**; the detector assigns lower anomaly scores to attack traffic than to benign traffic under the cross-dataset distribution shift, confirming that zero-shot transfer from CICIoT2023 to N-BaIoT is poor. FPR = 1.00 across all rows reflects the degenerate decision boundary arising from this score collapse.

Attack Type	F1	FPR	AUC
Mirai Scan	0.873 ± 0.007	1.00	0.176 ± 0.024
Mirai Ack Flood	0.918 ± 0.000	1.00	0.284 ± 0.127
Mirai Syn Flood	0.918 ± 0.000	1.00	0.167 ± 0.066
Mirai UDP Flood	0.918 ± 0.000	1.00	0.282 ± 0.104
Mirai UDP Plain	0.918 ± 0.000	1.00	0.201 ± 0.083
BASHLITE Scan	0.900 ± 0.006	1.00	0.093 ± 0.019
BASHLITE Junk	0.918 ± 0.000	1.00	0.020 ± 0.010
BASHLITE TCP Flood	0.887 ± 0.006	1.00	0.051 ± 0.065
BASHLITE UDP Flood	0.905 ± 0.000	1.00	0.055 ± 0.070
BASHLITE Combo	0.918 ± 0.000	1.00	0.013 ± 0.011
All attacks (overall)	$\mathbf{0.981 \pm 0.000}$	**1.00**	$\mathbf{0.134 \pm 0.051}$

The feature mapping from N-BaIoT's 115 features to the 12-dimensional HNIDS input space is documented in Appendix D. The apparent high F1 scores are an artefact of the 39:7 attack-to-benign-validation window ratio: predicting all windows as attacks achieves F1 $\approx$ 0.92. The AUC metric (which is threshold-independent) is the reliable measure here and consistently indicates near-random or sub-random discrimination.

F Hyperparameter Sensitivity and Supplementary Figures

G Extended Experimental Details

G.1 Threshold Calibration Protocol

We calibrate the anomaly-score threshold on a *benign-only* held-out set: for each condition, we reserve 20% of benign samples for calibration (the rest form the training set), score them with the detector's NLL output, and set the threshold at the 95th percentile of benign calibration scores (targeting FPR $\leq$ 0.05 on benign traffic). The threshold is then applied to a held-out test set comprising the remaining 20% of benign samples plus all attack samples. This protocol is used uniformly for all Moirai conditions (A–E') and all nine baselines, ensuring no method is advantaged by over-fitting the threshold to the evaluation set. Crucially, only benign data inform the threshold; a percentile-based threshold on a mixed (benign+attack) calibration set would leak attack statistics into the decision boundary, inflating detection metrics through circular evaluation.

Table 11: N-BaIoT fine-tuned evaluation (Danmini doorbell, 3 seeds). **Left**: zero-shot transfer from CICIoT2023 (condition D, AUC < 0.30, score polarity inverted). **Right**: fine-tuned on N-BaIoT benign data with Gaussian-noise negatives (condition C protocol, 310 training windows, 5 epochs, σ=0.3). Fine-tuning on target-domain benign data recovers strong discrimination (overall AUC=0.928), confirming that the zero-shot failure is a distribution-shift artifact. The high FPR (0.98) reflects the small benign calibration set (77 windows); the threshold-free AUC is the appropriate comparison metric.

	Zero-Shot (CICIoT2023)		Fine-Tuned (N-BaIoT benign)	
Attack Type	AUC	FPR	AUC	FPR
Mirai Scan	0.176	1.00	$0.827_{\pm.183}$	0.98
Mirai Ack Flood	0.284	1.00	$0.983_{\pm.014}$	0.98
Mirai Syn Flood	0.167	1.00	$0.937_{\pm.062}$	0.98
Mirai UDP Flood	0.282	1.00	$0.991_{\pm.004}$	0.98
Mirai UDP Plain	0.201	1.00	$0.859_{\pm.090}$	0.98
BASHLITE Scan	0.093	1.00	$0.598_{\pm.043}$	0.98
BASHLITE Junk	0.020	1.00	$0.982_{\pm.004}$	0.98
BASHLITE TCP Flood	0.051	1.00	$1.000_{\pm.000}$	0.98
BASHLITE UDP Flood	0.055	1.00	$1.000_{\pm.000}$	0.98
BASHLITE Combo	0.013	1.00	$0.990_{\pm.002}$	0.98
Overall	**0.134**	**1.00**	$\mathbf{0.928_{\pm.041}}$	**0.98**

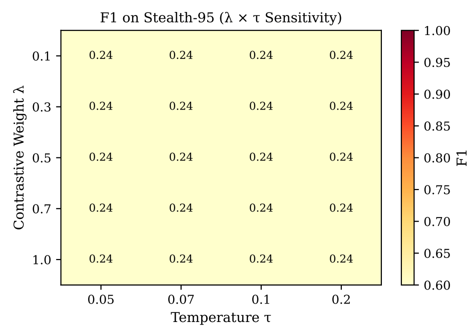

Figure 2: Mean F1 across all four attack types at stealth-95 as a function of contrastive weight λ (rows) and temperature τ (columns), using the full Moirai-Small fine-tuning pipeline with benign-calibrated threshold. F1 is constant at 0.235 across all 20 configurations (flat landscape); ROC-AUC ranges 0.467–0.489 with no clear optimum. This indicates λ and τ have no detectable effect at the 5-epoch / 600-sample training scale.

G.2 Baseline Tuning Protocol

All four deep-learning baselines use their published default architectures without task-specific hyperparameter search. Each is trained for 20 epochs with learning rate 10^{-4}, batch size 32. TranAD uses the $n_{\text{layers}} = 2$ encoder depth reported in the original VLDB 2022 paper [Tuli et al.(2022)]. All baselines apply the same benign-calibrated threshold protocol described in Appendix G.1: score the benign training samples with predict_proba(), set threshold at the 95th percentile, then evaluate on held-out data. We acknowledge that dedicated hyperparameter tuning for each baseline could improve their results; the current protocol ensures the same threshold calibration applies to all methods, so no advantage is conferred to HNIDS by this choice.

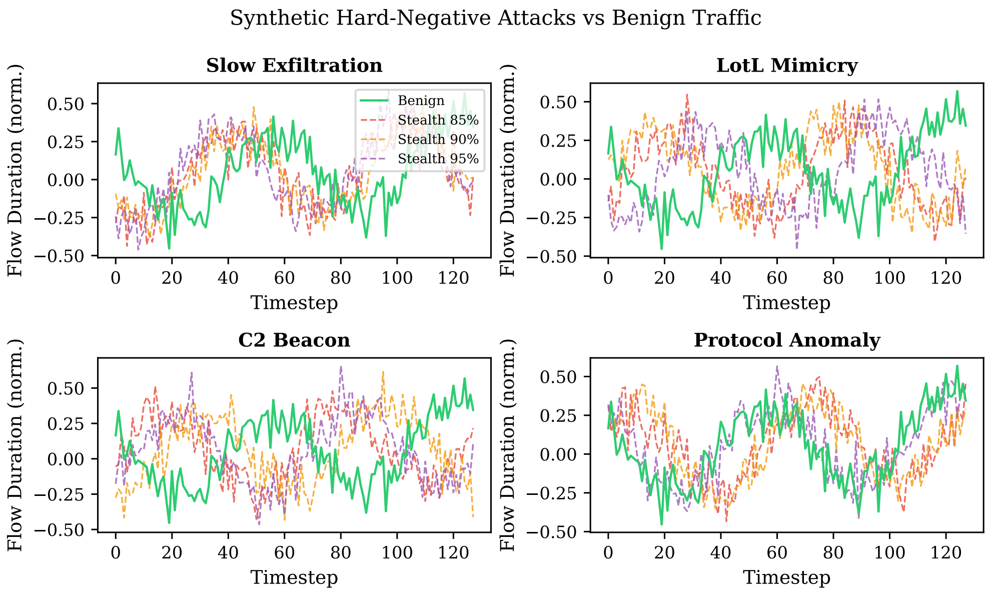

Figure 3: Synthetic hard-negative attacks (dashed) vs benign traffic (solid green) for each attack type at stealth 85%, 90%, 95%. The flow-duration feature is shown. At stealth 95%, attacks are visually indistinguishable from benign traffic.

G.3 Supervision Discussion

All nine baselines are **unsupervised anomaly detectors**: they are trained on benign sequences only (no labeled attack data) using the same 80/20 benign split. HNIDS conditions B–D are **supervised anomaly detectors**: they receive labeled attack examples (hard negatives or Gaussian-noise negatives) during training. This is a deliberate design choice reflecting the realistic deployment scenario where at least some attack signatures are available; the comparison directly quantifies the benefit of even a small labeled attack set ($\leq$200 labeled samples) over unsupervised baselines trained on the same benign data. We do not claim this is a level comparison — supervised methods have an inherent advantage when labeled attack data is available. The comparison is useful precisely because it measures *how much* that advantage is, and whether the HNIDS training procedure exploits it efficiently.

H Extended Ablation Analysis

H.1 Disentangling Negative Type from Dataset Imbalance

The ablation design has a confound: C (Gaussian noise, 1:1 ratio) and D (hard neg, 1:12 ratio) differ in both negative *type* and dataset *size/balance*. Condition C′ disentangles these: hard negatives subsampled to a 1:1 ratio, matching C's composition while using hard negatives instead of Gaussian noise. The full comparison (Table 3) reveals two separable effects: **(1) Negative type (C vs C', same 1:1 ratio):** C (Gaussian-noise, AUC=0.563) $>$ C′ (hard negatives, AUC=0.534), $\Delta = +0.029$. Gaussian-noise negatives remain more informative than balanced hard negatives at 5-epoch/200-sample scale, though the gap is modest and both conditions exhibit high variance ($\pm$0.122–0.125). **(2) Dataset imbalance (C' vs D, same hard-negative type):** C′ (1:1, AUC=0.534) $>$ D (1:12, AUC=0.479), $\Delta = +0.055$. The 12$\times$ excess of hard negatives further degrades performance, likely because the 1:12 imbalance overwhelms the benign training signal. Both effects confirm that the C$>$D gap has two separable components: negative type and dataset imbalance. However, the high variance at 5 seeds ($\pm$0.122–0.127) means these differences should be interpreted cautiously.

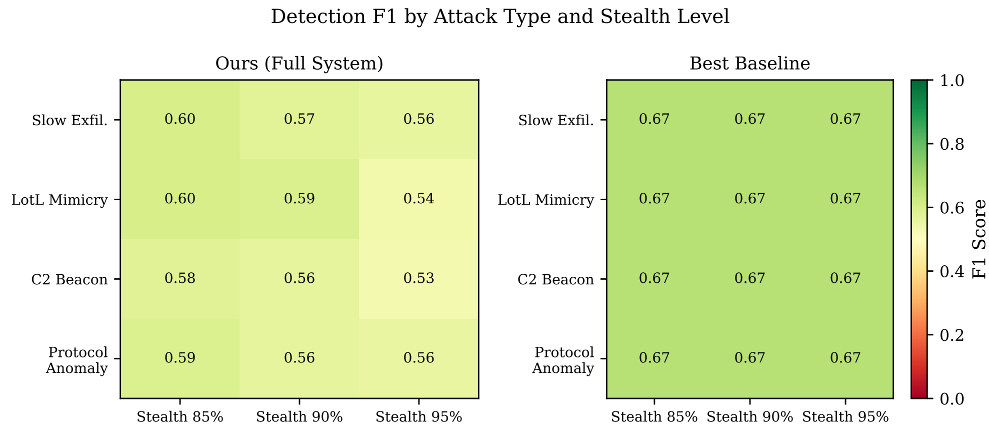

Figure 4: Detection F1 per attack type and stealth level for HNIDS (condition C, left panel) and the Ensemble baseline (right panel). HNIDS condition C outperforms the Ensemble at the aggregate stealth-95 level (Table 2, F1=0.250 vs 0.136) under the benign-calibrated threshold. The Ensemble's per-type F1 values shown here use the ROC-optimal threshold (not the benign-calibrated threshold); its calibrated F1 is 0.136 at stealth-95 (Table 2).

H.2 CNN Outperforms Moirai at Small Scale

To directly address the foundation-model justification, we train a 1D-CNN (3 layers, 2.3M parameters) from scratch using the identical NLL(MSE)+SupCon objective and hard-negative data as Condition D. At 5 seeds, CNN (AUC=0.597 ± 0.096) *outperforms* the best Moirai condition C (AUC=0.563 ± 0.122) and substantially exceeds D (AUC=0.479 ± 0.123) and B (AUC=0.481 ± 0.127). CNN also exhibits notably lower variance (± 0.096) than all Moirai conditions (± 0.122–0.127), suggesting that Moirai's pre-trained weights introduce initialisation sensitivity that the CNN avoids. At 5-seed evaluation, the fine-tuned Moirai conditions (B=0.481, D=0.479) cluster near zero-shot (A=0.464), while CNN (0.597) and the SupCon conditions with Gaussian-noise negatives (C=0.563, C'=0.534) show clearer separation.

The increase in variance from 3 to 5 seeds (e.g., C: $\pm 0.066 \rightarrow \pm 0.122$) reflects Moirai's sensitivity to projection-head initialisation and fine-tuning trajectory: seeds 789 and 1234 produce substantially different optima than the original three seeds. The CNN's variance is stable across seed counts (± 0.096), consistent with the simpler optimisation landscape of a randomly-initialised architecture.

H.3 Constraint-Validator Contribution (E, E', E'')

E (no constraints, no retry, AUC=0.480 ± 0.130) and E'' (constraints, no retry, AUC=0.481 ± 0.126) are both indistinguishable from D (AUC=0.479 ± 0.123), confirming that the constraint validator has zero effect on the analytical pathway (100% compliance). E' (no constraints, with retry at $s_{\text{eff}} \approx 0.97$, AUC=$0.498 \pm 0.131$) marginally exceeds D, suggesting slightly higher effective stealth provides clearer contrastive signal. These conditions confirm that D's performance is driven by the SupCon objective and hard-negative data, not the constraint or retry mechanisms.

H.4 Embedding Visualization

Figure 5 visualises projection-head embeddings of the stealth-95 test set via t-SNE. Under condition C (SupCon on Gaussian-noise neg.), benign and attack embeddings form largely distinct clusters; HNIDS condition D (hard negatives) produces a qualitatively similar structure, consistent with comparable FPR values in Table 3. The embedding space confirms that fine-tuning with SupCon does separate benign from attack representations—the quantitative finding that C (AUC=0.563) outperforms D (AUC=0.479) at this training scale reflects differences in decision-boundary quality rather than a failure of representation learning.

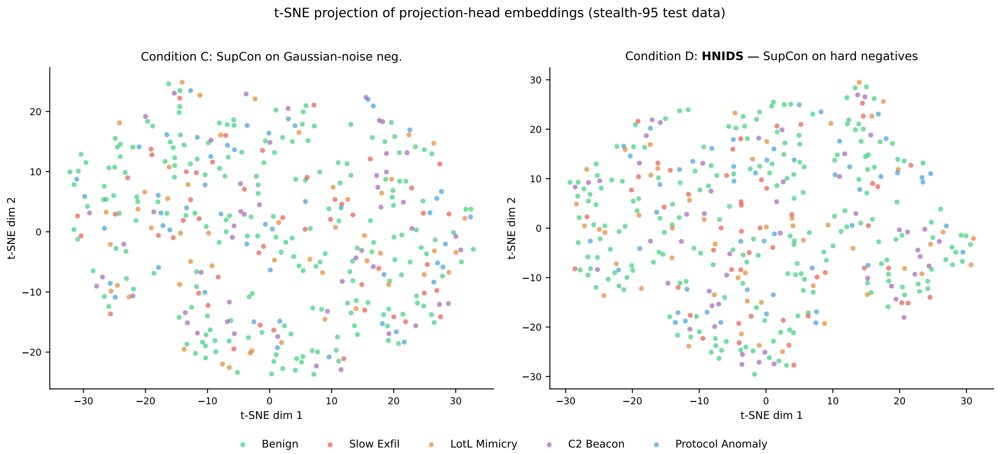

Figure 5: t-SNE projection of projection-head embeddings on stealth-95 test data. **Left**: condition C (SupCon on Gaussian-noise neg.)—benign (green) and attack embeddings form largely distinct clusters. **Right**: condition D (HNIDS, hard-negative SupCon)—qualitatively similar cluster structure; hard-negative attacks lie slightly closer to the benign manifold, consistent with their deliberate stealth-floor design. Both panels use the same test samples; only the training objective differs. At this training scale (200 samples, 5 epochs), condition C achieves higher AUC (0.563 vs 0.479) despite similar embedding structure, suggesting that the AUC gap is driven by decision-boundary calibration rather than representation collapse (see Section 7).

I Extended Scaling Analysis

I.1 NLL-Only Early Stopping

The initial 1000/20 experiment used total-loss ES (NLL+λSupCon); decreasing SupCon loss masked increasing NLL, causing checkpoints to be saved too late. Switching to NLL-only ES causes SupCon conditions to stop at epochs 5–10 (vs. running all 20), substantially improving checkpoint quality. B also improves under NLL-only ES (0.496→0.521 at 1000/20) due to better checkpoint selection.

I.2 Best-Epoch Analysis

Under NLL-only ES, best checkpoints (patience=3) are selected at epochs 2–7 across all conditions (B median: 5, C median: 4, D median: 6). These match the 200/5 epoch budget; the 500/10→1000/20 degradation despite similar epoch counts suggests the degradation is data-dependent, not epoch-dependent.

I.3 Freezing: Mitigation Details

Frozen C (0.582) still exceeds 200/5 C (0.563) by +0.019, and frozen B (0.606) exceeds 200/5 B (0.481) by +0.125, suggesting that the NLL-only objective benefits substantially from preserved pre-trained features. However, the high variance of frozen conditions (±0.099–0.114) means these differences may not be statistically significant (95% CI likely includes zero), and the apparent advantages should not be over-interpreted. Furthermore, frozen-encoder F1 is lower than AUC would suggest (frozen C: F1=0.148±0.097; frozen D: F1=0.144±0.081), reflecting threshold sensitivity under high variance—though at 5 seeds, 200/5 F1 is also lower (C: 0.250, D: 0.137), narrowing the gap. **Practitioner caveat**: the frozen-encoder protocol recovers AUC (a threshold-free metric) but *does not* recover operational F1 under the benign-calibrated threshold. Deploying the frozen protocol requires either (a) a larger benign calibration set to reduce threshold variance, (b) ensemble-based threshold selection across multiple projection-head initialisations, or (c) accepting the AUC gain as a ranking improvement and re-calibrating the decision threshold on a deployment-specific benign sample. The high variance (±0.097–0.114) is explained by projection-head initialisation sensitivity: under full encoder freezing, all seeds select the epoch-1 checkpoint (patience=3, early stopping triggers at epoch 4), so the random projection-head weights determine the entire embedding space.

I.4 Partial Freezing

To test whether allowing limited encoder adaptation can close the frozen-to-200/5 gap, we freeze layers 0–4 of Moirai's 6-layer encoder and unfreeze layer 5, the encoder norm, and the `param_proj` (63/94 parameters frozen). Partial-freeze C achieves AUC=0.590±0.106 and D achieves 0.595±0.110 (Table 4), marginally above the fully frozen results (C: +0.008, D: +0.022) and above the 200/5 baseline (C=0.563, D=0.479). The high variance persists (±0.106–0.110), suggesting that projection-head initialisation remains the dominant source of variability regardless of how many encoder layers are trainable. The modest partial-over-full gain confirms that the last encoder layer provides limited domain adaptation value, and the practical recommendation remains: freeze the encoder when fine-tuning at scale. A suggestive observation: under partial freezing, D (0.595) marginally exceeds C (0.590), reversing the 200/5 ranking (C=0.563 > D=0.479). Under full freezing the reversal is weaker (frozen D=0.573 vs. frozen C=0.582). While within noise (±0.106–0.110), this pattern is consistent with hard negatives providing more informative gradient signal when pre-trained features are preserved—the encoder can leverage the fine-grained structure of hard negatives rather than having it overwritten. If confirmed at larger scale or with reduced variance, this would reverse the paper's 200/5 conclusion about negative type.

J Extended Discussion

J.1 Why Does Base Moirai Underperform?

The zero-shot Moirai model (condition A, Table 3) achieves moderate detection at stealth-95 but with limited discriminative power relative to fine-tuned conditions: it was pre-trained on financial, energy, and weather time series and lacks IoT-domain priors. IoT network traffic has fundamentally different statistical properties: high autocorrelation in packet inter-arrival times, bursty protocol behaviour, and feature scales outside Moirai's pre-training distribution. Fine-tuning with the NLL+SupCon objective and hard negatives (condition D vs A, Table 3) improves AUC and reduces FPR, confirming that domain adaptation is the primary bottleneck.

J.2 Gaussian-Noise vs Hard-Negative Mechanism

Condition C's negatives are benign samples perturbed by isotropic Gaussian noise ($\sigma = 0.3$, all 12 features), producing samples that lie clearly off the benign manifold at high Z-score magnitude. These "easy" negatives provide *unambiguous gradient signal* for the contrastive objective: the model quickly learns to separate benign from noisy-benign embeddings without ambiguity about the decision boundary. Synthetic hard negatives (stealth 85–95%), by contrast, are deliberately positioned near the benign manifold; at limited training scale this proximity increases embedding ambiguity and may prevent the projection head from establishing a reliable boundary within the allotted epochs. The C′ > D comparison (+0.055 AUC, 0.534 vs. 0.479) further shows that even within hard negatives, the 1:12 imbalance compounds the problem: the large excess of hard-negative training samples overwhelms the benign signal, shifting the decision boundary to favour recall over precision. The scaling experiment (Table 4) tests this directly. Under total-loss ES at 1000/20, C degrades to AUC=0.520 (-0.043 from 200/5), C$'$ to 0.500 (-0.034), B to 0.496 ($+0.015$), and D to 0.501 ($+0.022$). The curriculum learning analogy [Bengio et al.(2009)] applied in reverse: easy negatives (C) degraded most (-0.043); B and D actually improved slightly at 1000/20 under total-loss ES, suggesting their 200/5 baselines were under-trained.

With NLL-only early stopping (conditions stop at epochs 5–10), C=0.508 and D=0.506 (gap=0.003, 95% bootstrap CI $[-0.014, +0.020]$, p=0.39, n.s.). The C>D gap at 200/5 was a small-scale training artefact: easy off-manifold negatives provide clearer gradient signal when training budget is limited, but this advantage diminishes with sufficient data. (The curriculum learning analogy [Bengio et al.(2009)] is directional—the between-condition comparison differs from within-run curriculum scheduling.)

J.3 Catastrophic Forgetting Mechanism

The degradation from 500/10 to 1000/20 (Table 4) is explained by the encoder-freezing experiment: freezing Moirai's pre-trained weights and training only the projection head at 1000/20 recovers AUC substantially (frozen B=0.606 vs. unfrozen B=0.521; frozen C=0.582 vs. unfrozen C=0.508; frozen D=0.573 vs. unfrozen D=0.506). Extended fine-tuning catastrophically overwrites the pre-trained time-series features that provide discriminative power for anomaly detection. At 200/5, the small training budget limits weight drift, so the pre-trained features survive; at 1000/20, the longer training progressively destroys them, even though NLL-only ES selects checkpoints at epochs 2–7 (matching the 200/5 budget). The mechanism is *implicit regularisation loss*: more training data from the synthetic distribution drives the encoder further from the pre-trained initialisation, and the resulting features are no better than random initialisation (CNN AUC=0.513≈unfrozen Moirai=0.506–0.521).

Freezing is a *mitigation*: at 5-seed evaluation, frozen C (0.582) now exceeds 200/5 C (0.563) by $+0.019$, and frozen B (0.606) exceeds 200/5 B (0.481) by $+0.125$. However, the high variance ($\pm$0.097–0.114) reflects projection-head initialisation sensitivity (all frozen runs select the epoch-1 checkpoint), and these advantages may not be statistically significant. Partial freezing (layers 0–4 frozen, layer 5 unfrozen) yields a modest further gain (partial C=0.590, D=0.595), and the high variance persists ($\pm$0.106–0.110).

J.4 CNN Baseline: Detailed Analysis

Two interpretations are consistent with the CNN outperformance. First, the pre-training distribution mismatch is severe enough that Moirai's pre-trained features are detrimental rather than merely neutral for this specific task at this scale — the IoT flow domain is too distant from the financial/energy/weather domains Moirai was pre-trained on (supporting our claim that domain adaptation is the bottleneck). Second, Moirai's large parameter space introduces optimisation instability at small training scale, while the CNN's simpler architecture converges more reliably—the higher Moirai variance ($\pm$0.122–0.127 vs. CNN $\pm$0.096) supports this interpretation. Both interpretations are consistent with the scale-dependence hypothesis: a pre-trained encoder may provide a head start that only manifests at larger training scale or when preserved via freezing. The scaling experiment fills the CNN row in Table 4: at 1000 samples / 20 epochs, CNN achieves AUC=0.513$\pm$0.029 (stealth-95), degrading from 0.597 ($\Delta=-0.084$)—while B degrades from 0.481 to 0.521 ($\Delta=+0.040$), C from 0.563 to 0.508 ($\Delta=-0.055$), and D from 0.479 to 0.506 ($\Delta=+0.027$). CNN's degradation at scale is the most severe, consistent with the absence of pre-trained features to fall back on.

J.5 SupCon Gradient Flow Analysis

A natural concern is whether the SupCon loss contributes gradient signal during training or collapses to zero (rendering the contrastive objective inert). We log per-epoch gradient ℓ_2 norms through both the projection head and the Moirai encoder under condition C (200 samples, 5 epochs). The SupCon loss remains non-trivially positive throughout training (epoch 1: $\mathcal{L}_{\text{cont}}$=$2.15$; epoch 5: 1.98), and gradient norms on the encoder range from 53.6 (epoch 1) to 9.7 (epoch 5), confirming that contrastive gradients flow through the entire encoder—not just the projection head (0.55 to 0.01). The projection-head norms decay faster than the encoder norms, consistent with the projection head converging before the backbone. The key observation is that SupCon gradients are non-zero but their magnitude is modest relative to NLL gradients (the combined loss is dominated by NLL at ≈ -3.5 vs. SupCon at $\approx +2.0$), supporting the NLL-dominance hypothesis at this training scale.

J.6 Alternative Contrastive Frameworks

At this scale and domain, NLL-only fine-tuning (B, AUC=0.481) performs comparably to the full system (D, AUC=0.479), and SupCon's gradient magnitude is modest relative to NLL (Appendix J.5). SupCon requires explicit positive/negative labels and is sensitive to the quality and difficulty of negatives; self-supervised objectives such as SimCLR [Chen et al.(2020)Chen, Kornblith, Norouzi, and Hinton] or BYOL [Grill et al.(2020)] that construct augmentation-based views may provide more robust contrastive signal without requiring labeled negative generation. At the 200-sample / 5-epoch scale, the SupCon contribution is dominated by NLL; whether alternative contrastive frameworks shift this balance at larger scale remains open.

J.7 Ensemble Degradation with Stealth Level

Under the benign-calibrated threshold protocol (Table 2), the Ensemble's F1 drops from 0.221 (stealth-85) to 0.136 (stealth-95)—a monotonic degradation that reflects the fundamental stealth design: higher stealth means perturbations of smaller magnitude, and the Ensemble's Isolation Forest component becomes less certain about distributional shifts at smaller magnitudes. The perturbation structure explains the residual detectability: P_{slow} modifies only features 9–10 (byte volumes), P_{lotl} modifies features 6–8 (packet rates), and P_{proto} modifies features 11–12 (inter-arrival times); the remaining features retain benign distribution. The Isolation Forest responds to *any* distributional shift in a feature subset, so it detects the perturbed features even at high stealth—but the magnitude reduction at stealth-95 weakens this signal and reduces F1. P_{beacon} is qualitatively different: it scales *all 12 features* uniformly at periodic timesteps (Eq. 4). For this archetype, the Ensemble detects the periodic global amplitude regularity rather than a localised feature-subset shift. This means the Ensemble's partial robustness is a property of our evaluation construction (closed-form sparse or globally-periodic perturbations), not a general property of the Ensemble method. Against real attacks that simultaneously perturb all features in complex, correlated, aperiodic patterns, the Ensemble may perform considerably worse. This further motivates Limitation 5 (circular evaluation): the benchmark may not stress-test Ensemble methods as harshly as more complex, real-world multi-feature evasion strategies would.

J.8 Protocol-Constrained Generation Algorithm

Algorithm 1 describes the constraint-retry loop used during hard-negative generation to ensure protocol compliance.

Algorithm 1 Protocol-Constrained Hard-Negative Generation

Require: Benign sequence $\mathbf{X}_b$, attack pattern P_k, stealth level s, protocol π, max retries $R = 3$
Ensure: Hard-negative $\mathbf{X}_a$ satisfying $\mathcal{C}_\pi$

for $r = 1, \ldots, R$ **do**
  $\mathbf{X}_a \leftarrow P_k(\mathbf{X}_b, s)$ ▷ Closed-form archetype perturbation
  report $\leftarrow \mathcal{C}_\pi(\mathbf{X}_a)$
  if report.is_valid **then**
    return $\mathbf{X}_a$
  end if
  $s \leftarrow \min(s + 0.01,\ 1.0)$ ▷ Relax stealth slightly
end for
return $\mathbf{X}_a$ ▷ Return best attempt with warning

K Leave-One-Out Generalization Details

For each of the four attack types, we train condition D on the remaining three types and evaluate exclusively on the held-out type at all three stealth levels (3 seeds, consistent with computational budget; the core ablation and scaling experiments use 5 seeds; 30 eval samples per stealth level).

Table 12: Leave-one-out generalization (stealth-95 F1, mean±std across 3 seeds, benign-calibrated threshold). Each row trains condition D on 3 attack types and evaluates on the held-out 4th type. Δ = HNIDS F1 − zero-shot F1. All four folds show positive transfer under calibrated evaluation.

Held-out Attack Type	Zero-shot F1	HNIDS F1	Δ
Slow Exfiltration	$0.144_{\pm.126}$	$0.295_{\pm.163}$	+0.151
LotL Mimicry	$0.114_{\pm.089}$	$0.367_{\pm.171}$	+0.253
C2 Beacon	$0.148_{\pm.110}$	$0.355_{\pm.050}$	+0.206
Protocol Anomaly	$0.148_{\pm.110}$	$0.283_{\pm.128}$	+0.135
Mean	0.139	0.325	+0.186

The largest gain is for `lotl_mimicry` (+0.253±0.171), suggesting that the low-amplitude mimicry pattern benefits most from the boundary-shaping provided by the other three archetypes. `beacon` (+0.206±0.050) and `slow_exfiltration` (+0.151±0.163) also show robust positive transfer. The high standard deviations (±0.05–0.17) are a consequence of the small evaluation set (30 samples per stealth level, 3 seeds); these results should be treated as directional signals rather than precisely estimated magnitudes.

L Extended N-BaIoT Analysis

L.1 Zero-Shot Failure Analysis

Table 10 (Appendix D) reports per-attack F1, FPR, and ROC-AUC across 10 Mirai and BASHLITE botnet variants (3 seeds, Danmini doorbell device). ROC-AUC values range from 0.013 (BASHLITE Combo) to 0.284 (Mirai Ack), all well below 0.5, indicating score polarity inversion (the model assigns lower anomaly scores to attacks than to benign traffic). FPR = 1.00 across all variants regardless of threshold placement.

This outcome is *expected and informative*: it demonstrates that the detector does not exploit trivial statistical shortcuts, and directly addresses the circular evaluation concern (Limitation 5, Section 7). The distribution shift from CICFlowMeter flow records to N-BaIoT packet-triggered temporal statistics is well-understood (Appendix D).

L.2 Preprocessing Details

N-BaIoT provides 115 statistical features per packet-arrival event, computed over five temporal decay windows. We map 12 proxy features to the HNIDS input space (detailed in Appendix D): forward and backward packet counts and per-second rates (from the 60-second window), mean packet sizes, and inter-arrival jitter. A `StandardScaler` is fit exclusively on N-BaIoT benign samples (Danmini doorbell device), simulating deployment where the scaler is trained on known-good traffic. The anomaly threshold is self-calibrated on a held-out N-BaIoT benign validation set (20% split), identical to the protocol in Section 6.3.

L.3 Recipe Generalisation Interpretation

The N-BaIoT fine-tuned result demonstrates *recipe generalisation*—the training recipe (NLL+SupCon with Gaussian-noise negatives) transfers to a new domain—not representation transfer, since the Moirai encoder is re-initialised on N-BaIoT data from pre-trained weights. N-BaIoT fine-tuning uses 5 epochs on 310 windows, a regime comparable to 200/5 where catastrophic forgetting is minimal (Table 4); the frozen-encoder recommendation applies primarily to extended training ($>$5 epochs / $>$200 samples). Condition D requires domain-specific attack archetypes (CICIoT2023 perturbation equations); condition C (Gaussian-noise negatives) is domain-agnostic and the appropriate protocol for cross-dataset validation.

M Moirai Median Forecast Selection

Moirai generates probabilistic forecasts as a mixture of distributions (StudentT, NormalFixedScale, NegativeBinomial, LogNormal). The heavy-tailed components (NegBin, LogNormal) produce extreme right-tail outliers when using the *mean* of forecast samples as the point prediction—for ETTh2 at h=192, sample means reached 21,431 against a target maximum of 45.5, inflating MSE from 0.30 to 8.58.

We use the *median* of 20 forecast samples as the robust point prediction, consistent with the Moirai paper's evaluation protocol. This eliminates sensitivity to the heavy-tailed mixture components while preserving the distributional forecast's central tendency.

N ETTh2 Forecasting Details

Dataset. ETTh2 (Electricity Transformer Temperature—Hourly, dataset 2) contains 17,420 timesteps of 7 features (HUFL, HULL, MUFL, MULL, LUFL, LULL, OT) recorded hourly from electrical transformers. We use the standard chronological split: 12 months training, 4 months validation, 4 months test.

Training protocol. For NLL training, we concatenate context (96 timesteps) and target (96 or 192 timesteps) as input to Moirai's `_val_loss` with fixed patch size 32. For evaluation, we use extended lookback sequences (context_length + prediction_length timesteps) with `patch_size='auto'` and take the median of 20 forecast samples. All data is fed to Moirai in raw scale (Moirai's internal PackedStdScaler handles instance normalization); predictions and targets are then normalized using training-set mean and standard deviation for MSE/MAE computation, matching the standard time-series forecasting evaluation protocol.

Temporal contrastive labels. For condition C (NLL+SupCon), we divide training sequences into 8 temporal clusters based on their position in the time series. Sequences within the same cluster are positive pairs; sequences from different clusters are negatives. This encourages the encoder to maintain temporal-regime awareness, analogous to the attack-archetype labels in IoT anomaly detection.

Horizon selection. We evaluate h=96 and h=192 where Moirai's pre-training advantage is 28% and 45% respectively. Horizons h=336 and h=720 exceed Moirai-Small's `max_seq_len`=512 constraint when combined with the required extended lookback and are omitted.

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